The Evolution of Transposable Elements in Trichinella Nativa & Spiralis

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TRICHINELLA

- Genus of parasitic roundworms (phylum is *Nematoda*)
- Found in infected mammals, birds, reptiles [1, 2]
- Can cause an infection called trichinellosis in humans if they eat raw/undercooked meat from animals that are infected [2]
- Trichinella is an important genus to study because it effects a wide range of animals and can remain undetected which can lead to infection [2]

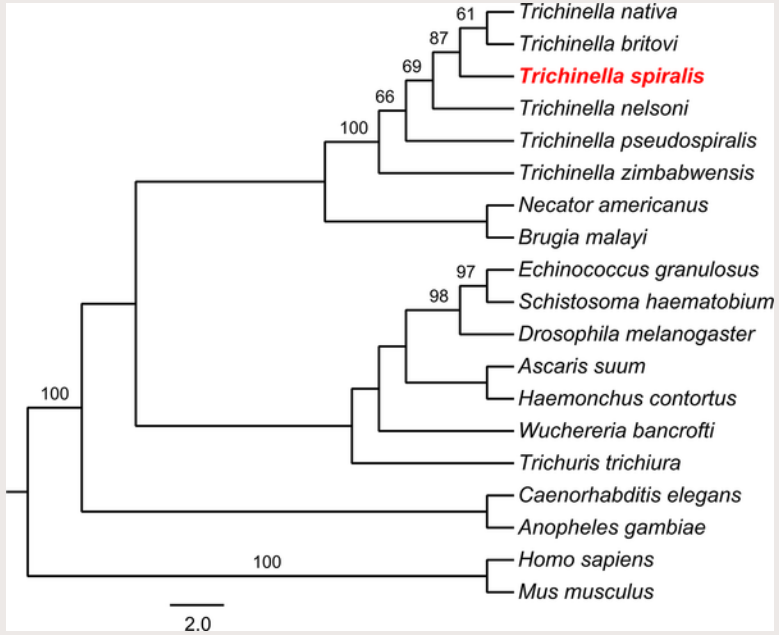
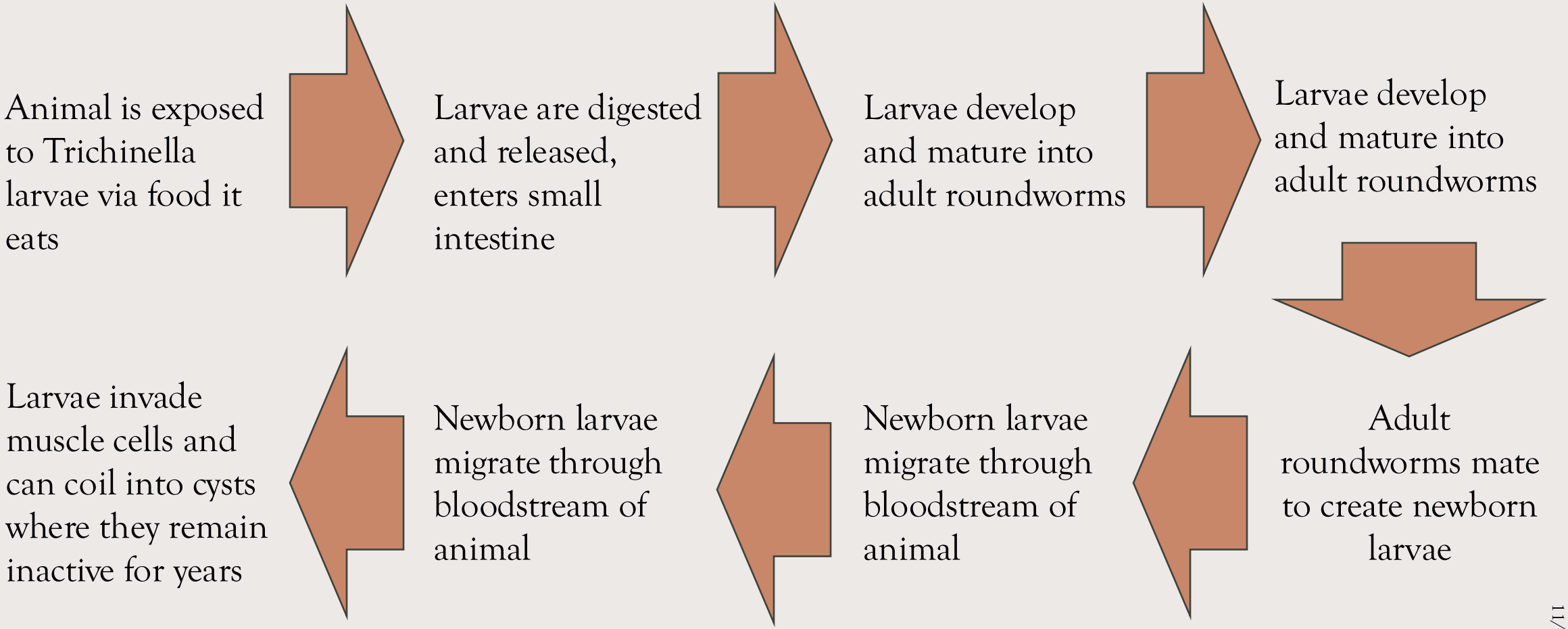


Image obtained from [1] Sun et al.

TRICHINELLA INFECTIONS CAN REMAIN UNDETECTED FOR YEARS [3]



TRICHINELLA SPIRALIS IS NOT FREEZE RESISTANT

- Typically found in animals that are either human dependant/live closely with humans or share a close genetic or evolutionary relationship with humans [2]
- Examples of some animals include pigs, rats, horses, boars, dogs and cats [2]
- Found worldwide, but commonly in Europe, USA, Asia and West Africa [2]
- Larvae in infected animals are killed at temperatures of -20 degrees Celsius (not freeze resistant) [2]



Image obtained from https://www.invasivespeciescentre.ca/invasive-species/meet-the-species/land-and-animal-invertebrates/wild-pigs/



Image obtained from https://shropshiremammalgroup.com/shropshire-mammal-atlas-2/wild-boar/

TRICHINELLA NATIVA IS FREEZE RESISTANT

- Typically found in animals that are common in the Artic [2]
- Some examples of animals include polar bears, arctic foxes, wolves, walruses, boar [2]
- Common in Artic and subarctic regions of North America, Europe and Asia [2, 3]
- These regions are where T. nativa is most common because of their colder climate [2]
- Larvae from T. nativa are freeze resistant so they can survive for many years in cold temperatures [2]
- T. nativa is basically immune to death at lower temperatures and researchers have not found a temperature at which it can be killed [2]



Image obtained from https://kids.nationalgeographic.com/animals/mammals/facts/arctic-fox



Image obtained from https://en.wikipedia.org/wiki/Polar_bear

THE EVOLUTION OF TRANSPOSABLE ELEMENTS AND TRICHINELLA

- Transposable elements are known as "jumping genes"
- Important in shaping gene evolution – we can use TEs to track how T. spiralis and nativa have changed over time [1]
- TEs can help us keep track of what parts of T. spiralis and T. nativa are independently evolved and which ones are a shared ancestor [1, 2]
- Evolution of these TEs and Trichinella is what has allowed T. nativa to diverge and become freeze resistant while T. spiralis is not [2]

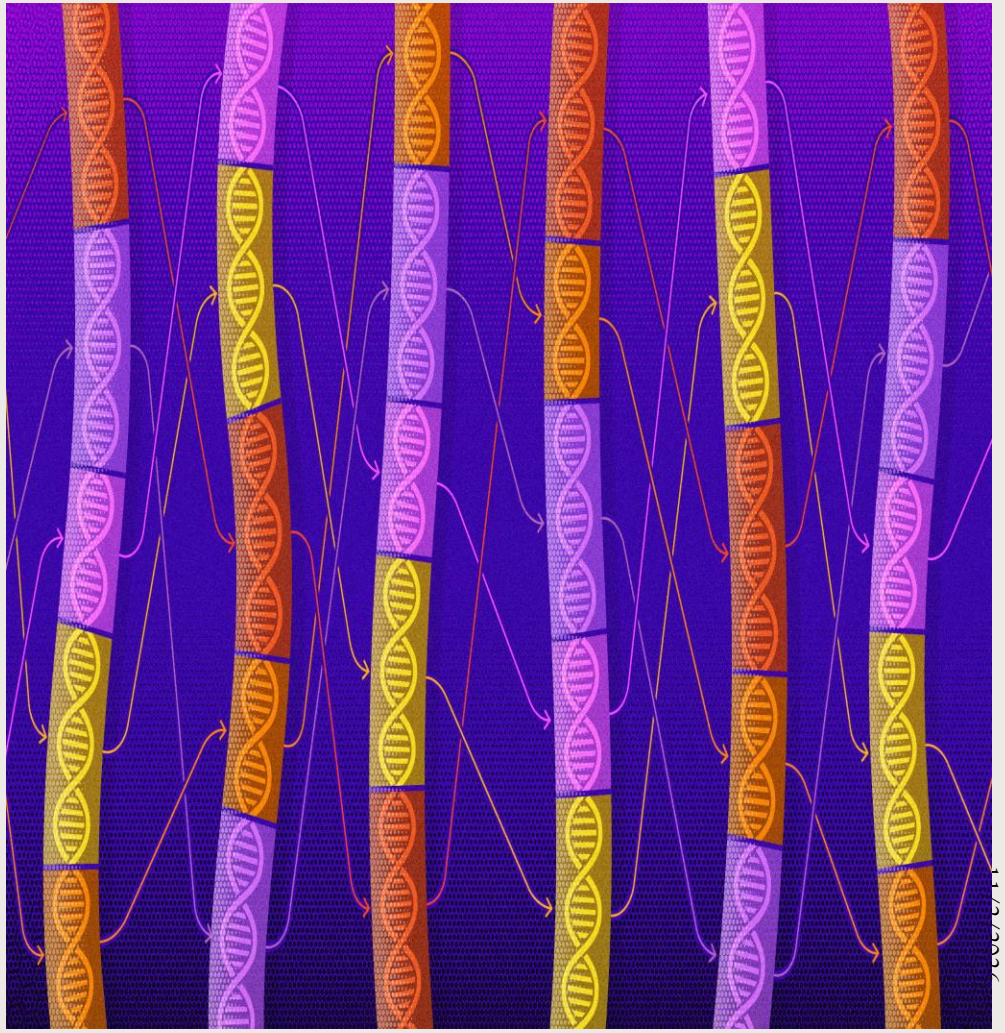


Image obtained from https://www.quantamagazine.org/scientists-catch-jumping-genes-rewiring-genomes-20210512/

NATIVA AND SPIRALIS HAVE DIFFERENT DNA TRANSPOSONS & RNA RETROSPOSONS

- Transposable elements have DNA transposons and RNA retroposons, both of which are different classes of TEs and can vary between T. nativa and T. spiralis [1, 3]
- Tracking the differences in DNA transposons and RNA retroposons between T. nativa and T. spiralis is another way for us to keep track of the evolution and divergence of the two species from their common ancestor [3]
- Based on this, we investigated the following: ***Do T. spiralis and T. nativa differ in the timing of transposable element activity and has either species had a recent TE expansion since they split?***

```
# Groups:   matching_class [11]
  matching_class        n
  <chr>             <int>
 1 DNA/TcMar-Tc1         81
 2 DNA/PIF-Harbinger      6
 3 DNA/MULE-MuDR          5
 4 DNA/TcMar-ISRm11       3
 5 DNA/TcMar-Tc4          3
 6 DNA/TcMar-Mariner      2
 7 DNA/hAT-Ac             2
 8 DNA/TcMar-Pogo         1
 9 DNA/TcMar-Tc2          1
10 DNA/hAT                1
11 DNA/hAT-Tip100         1
```

```
   matching_class       n
   <chr>            <int>
 1 PLE                349
 2 PLE/Naiad          345
 3 LTR/Gypsy          210
 4 LTR/Pao            202
 5 LINE/CR1            84
 6 LTR/Ngaro           14
 7 LINE                 2
 8 LINE/RTE-BovB        2
 9 SINE/tRNA-Deu-L2     2
10 LINE/I-Jockey        1
11 LTR/DIRS             1
12 SINE/tRNA-RTE        1
```

Images outline names of DNA transposons (left) and RNA retroposons (right) for T. nativa

METHODS – R CODE/PIPELINE

#	*Step*	*Command / Functions*	*Output*
1	Import	read.csv()	Repeat tables, one per species
2	Clean	mutate(), str_replace_all()	Parenthesis around values/numbers
3	Keep TEs only	filter(%notin%, %in%)	Non-TE repeats removed
4	Dense contigs	group_by(), summarise(), filter(), count(), log10()	Contigs with > 50 repeats (Inflection point, log10): 268/220
5	Label & combine	mutate(if_else()), bind_rows()	1661 TE copies: DNA vs Retro, by species
6	Checks	geom_histogram()	No impossible values; irregular distributions > non-parametric test
7	Analysis / Statistical analysis	wilcox.test() *4, prcomp(), geom_smooth()	Age (Fig 1), length (Fig 2), composition (Fig 3)
8	Report	ggplot(), ggsave()	Figures with medians, p-values, n

METHODS – R CODE/PIPELINE

- **Divergence from reference sequence was used as a relative age proxy. [4]**
- **Repeat counts per contig and TE copy lengths were log10- transformed, because both are strongly right-skewed (a few very large values); this made the inflection point visible and spread-out short copies for plotting.**

STATISTICAL PIPELINE:

Wilcoxon rank-sum tests	**PCA**
Q: Does TE age differ between TE types and species?	Q: Does TE composition differ between species?
Sample: Irregular distributions, outliers, unequal n (85-100 vs 738)	Input: contig x TE family count (23 families; absent=0)
Test (x4): DNA vs retro within species; species vs species within TE type	Method: scaled counts, prcomp()
Reported: medians, p-values; Bonferroni threshold $p<0.0125$	Output: PC1 vs PC2 by species, 95% ellipses; loadings = families driving each axis

FINDINGS

TE timing is largely shared between the species, with no recent expansion in either.
Violin show the distribution, boxes the median and interquartile range and points individual TE copies. Median and Wilcoxon rank-sum in the figure.

- Retrotransposons: same timing in both species (medians 21.3% vs 21.4%; p = 0.76).
- *T.nativa*: DNA transposons and retrotransposons have the same age (21.2% vs 21.3%; p = 0.48).
- All distribution peak at ~20-25%, with no bulge at low divergence: **no recent burst in either species.**

Fig 1: Divergence of TE copies from their reference sequences, by TE type and species.



DNA transposons: slightly older in *T.spiralis* than in *T.nativa* (22.8% vs 21.2%; p = 0.030), and than retrotransposons within *T.spiralis* (p = 0.038), but neither survives correction for four tests (threshold p < 0.0125), so this is tentative trend.

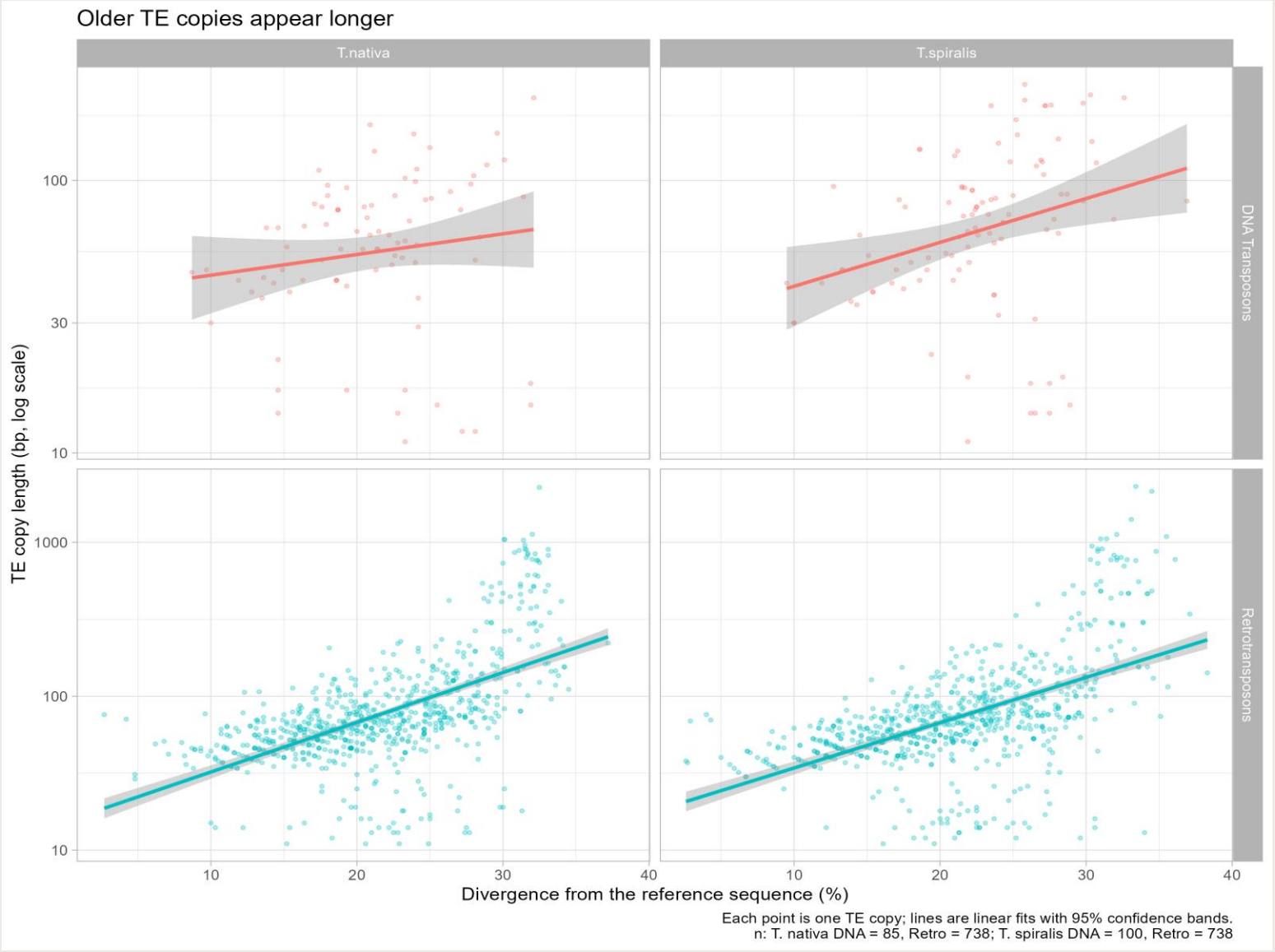
FINDINGS

TE copies are short, old fragments: no sign of a recent expansion.

- Most TE copies are under ~200 bp, far shorter than full length TEs (often several kb): **remnants of old insertions.**
- No young, near full-length copies in either species, **consistent with no recent TE expansion**.
- Unexpectedly, length increases with divergence, clearly for retrotransposona (n= 738 per species), weak for DNA transposons (wide confidence band n = 85 & 100).
- The same pattern in both species.

Old short fragments may go undetected, so only longer old copies remain.

Fig 2: TE copy length versus divergence from the reference sequence, by TE type and species



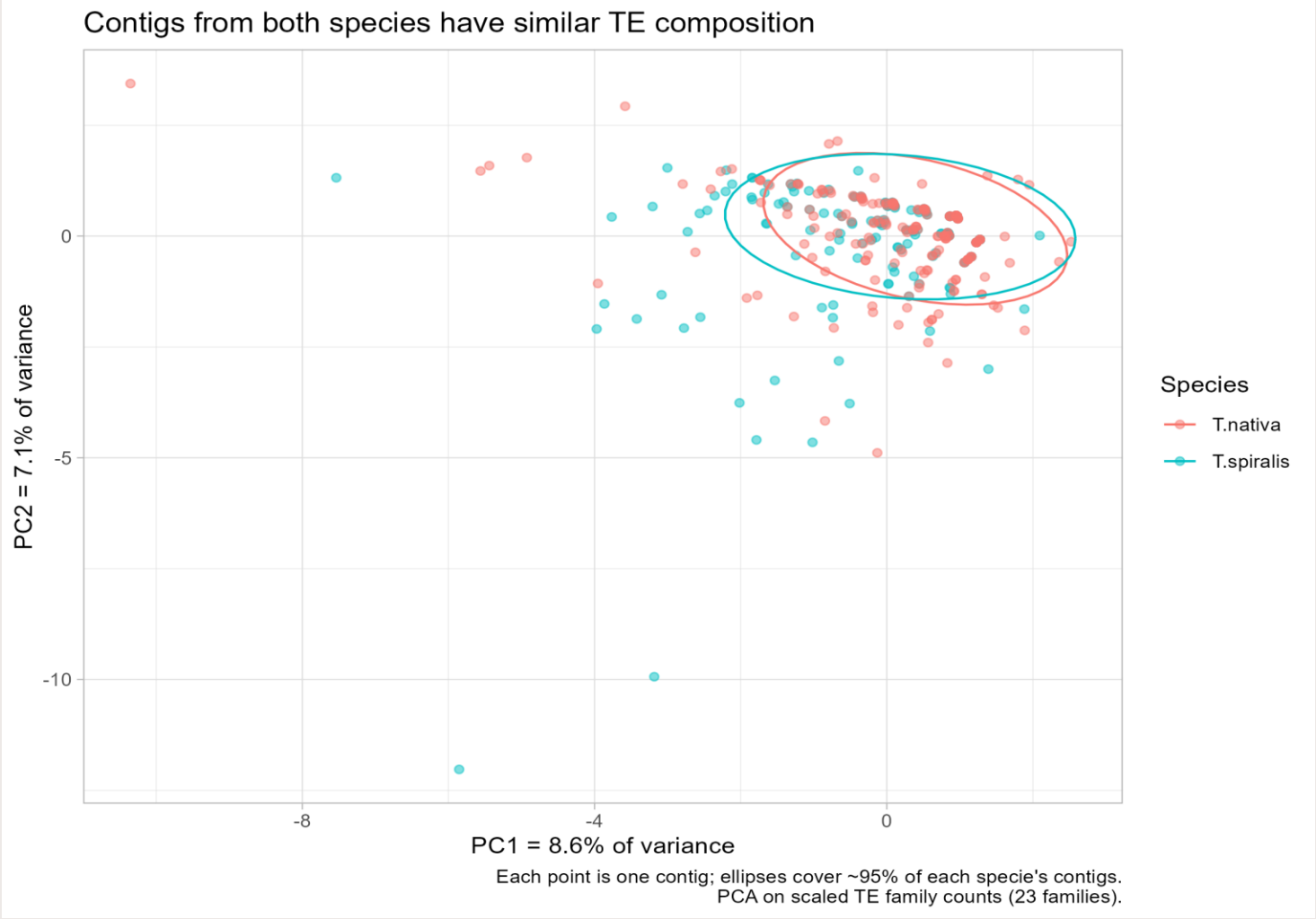
FINDINGS

TE family composition is largely shared: no family expanded in only on species.

- The species' contigs overlap almost completely, with no clustering.
- PC1 (8.6%) is driven by PLE, PLE/Naiad, DNA/TcMar-Tc1 and LINE/CR1; PC2 (7.1%) by LTR/Pao, LTR/Gypsy and DNA/TcMar-Mariner.
- A few outlier contigs are rich in particular families (e.g. LTR-rich *T.spiralis* contigs).

PC1 + PC2 explain only 15.7% of variation, so composition may be spread across many weaker axes.

Fig 3: Principal component analysis (PCA) of TE family composition across repeat-dense contigs in *T.nativa* and *T.spiralis*



If a TE family had expanded in one species after the split; that species' contigs would carry more of it and separate in the PCA. No evidence of lineage-specific expansion.

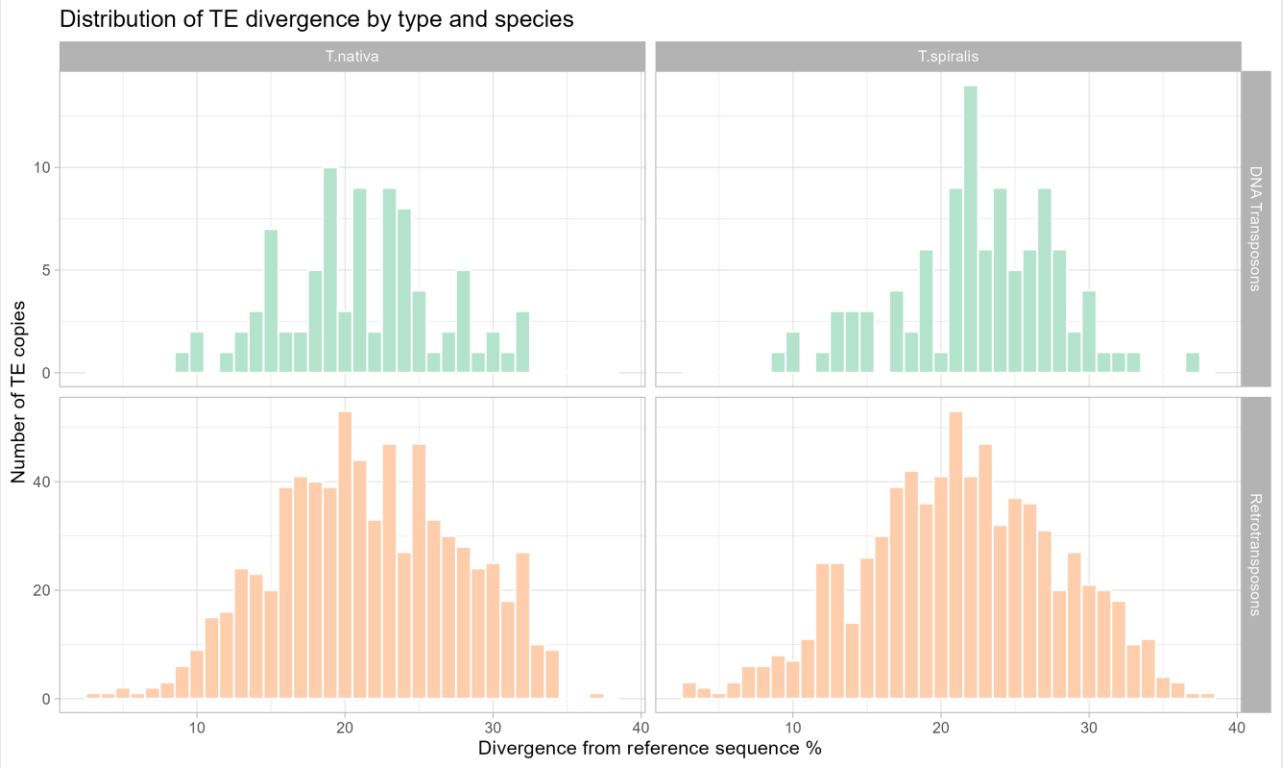
RESULTS & DISCUSSION

The Distribution Histograms showed an irregular distribution of *T.spiralis'* DNA transposon values, concentrated around 21-27%.

The Distribution Histograms showed patchy distribution of *T.nativa's* DNA transposon values.

Retrotransposons were roughly bell – shaped in both species.

Because some groups are not normally distributed and group sizes were very unequal (85-100 vs 738), we used the Wilcoxon Ranked Sum test because it allowed us to compare the ranks of values without assumming normality.



We looked at the data's pattern before deciding the test.

RESULTS & DISCUSSION

TE History is largely shared between the *T.nativa* and *T.spiralis* species, which went against our expectations.

- **Timing:** we expected differences between species. We found the same timing, with only DNA TEs slightly older in *T.spiralis* (tentative).

>>> Possible explanation: most TE activity that likely happened before the species split.

- **Recent expansion:** we expected one in at least one species. We found no peak of young copies.

>>> Possible explanation: TEs may now be inactive, or young copies are missing from the draft assembly.

- **Copy length:** we expected older copies to be shorter (> deletions and mutations). We found they appear longer.

>>> Possible explanation: Detection bias. Old, short fragments are too diverged for RepeatMasker to find. [6]

- Composition (PCA): we expected the species to separate. There was extensive overlap in contigs composed of similar TE families.

>>> **Possible explanation:** TE families were inherited from the common ancestor.

RESULTS & DISCUSSION

The data supports the main conclusion, but is too small for finer differences.

Dataset:

- Retrotransposons (n = 738 per species): Large, with clear and consistent results.
- DNA transposons (n = 85, 100): Small, wide uncertainty, and the p-values don't survive correction for four tests.
- PCA: PC1 + PC2 explain only 15.7% of the variation.

Retrotransposons outnumber DNA transposons about 8 to 1 in both species:

Retrotransposons copy and paste (original stays and a new copy is added), while DNA transposons mostly cut and paste (they move but don’t multiply directly). Over time, copy-and-paste elements build up far more copies.

PLE + PLE/Naiad are the most abundant classes in both species (390 in *nativa*, 370 in *spiralis*), a distinct type of retrotransposon that is less studied. [7]

CONCLUSION

***T.spiralis* and *T.nativa* share an old TE history, with no recent expansion in either species.**

- Timing: TE activity looks largely the same in both species. Retrotransposons are identical in age (21.3% vs 21.4%), and DNA transposons are only slightly older in *T.spiralis* (Fig.1).
- Recent expansion: none detected. There's no low-divergence peak (Fig. 1), no young full-length copies (Fig. 2), and no TE family enriched in one species (Fig. 3).

- Interpretation: most TE activity probably happened before the species split, and both genomes now carry decayed remnants of a shared TE history.

LIMITATIONS

- Divergence used as a proxy for age, it gives relative age, not absolute age. [4]
- Young TE copies are hard to assemble, so some maybe be missing from these draft genomes. [5]
- Only repeat-dense contigs (>50 repeats) were analyzed, and DNA transposons sample are small (n = 85, 100).

Future directions:

- Convert divergence into time, using mutation rate and the species' split date.
- Use more complete (long-read) assemblies, to recover young TE copies.
- Use RNA-seq and proteomics to test whether any TEs are still transcribed or produce transposition proteins, which genomic data alone cannot show.

REFERENCES

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WHAT DID WE ASK COPILOT?

SUMAYYAH KHAN

- The names of the list of names from *matching_class* were inputted into co-pilot, and it was asked to **"sort the list into transposable and non-transposable elements from above"**
- The list of only transposable elements was again inserted into copilot and it was asked to **"sort the list into DNA transposons and RNA retroposons"**
- The above was done for both T. spiralis data and T. nativa data
- Used Co-pilot for spell-check and grammar.

Prompts:

Inputted list of TEs and non TEs from matching_class into copilot

Sort the above list into transposable and non-transposable elements

Sort the above list into DNA transposons and RNA retroposons

WHAT DID WE ASK COPILOT?

SHARIQ HASHMI

Prompts:

- **Explain T.nativa species and how it compares with spiralis genotype?**
- **What makes them cold resistant?**
- **how to TEs between the two differ and interact?**
- **[1] "PLE" "PLE/Naiad" "LTR/Pao" "LTR/Gypsy" "LINE/CR1" ... Can you create a list Non-TEs? por favor**
- **Just the list of classes that arent TEs**
- **"Simple_repeat" "Low_complexity" "PLE" "Unknown" "PLE/Naiad" ... "ARTEFACT" this list**
- **"Simple_repeat" "Low_complexity" "Unknown" "PLE" "PLE/Naiad" ... "DNA/TcMar-ISRm11"**
- **Give me a list of prompt from this chat**
- **"How to interpret wilcoxon ranked sum results."**
- **"Principal component analysis, for examinig variation in contig composition based on TE family. How to understand the results.**
- **"What is the major difference between between T. Sparalis and T. Nativa in DNA adaptions"**
- **Give me a list of prompt from this chat**
- **Co-Pilot was used for grammar and spell check.**

WHAT DID WE ASK COPILOT?

Jordyn Anene

- Background Information on species differences
- Communication Help
- Used Co-pilot for spell-check and grammar.
- Effective statistical analysis

Prompts:

- **"What is the major difference between between T. Sparalis and T. Nativa in DNA adaptions"**
- **"What makes Nativa more cold resistant"**
- **"Is the Wilcox Ranked Sum test the most effective?"**
- **"How to explain politely to group members we need to explain the Wilcox test"**

WHAT DID WE ASK COPILOT?
LUIS VILLASANA

Copilot:

- from this list which ones are dna transponsons and retrotransponsons "Simple_repeat" "Low_complexity" "Unknown" [4] "PLE" "PLE/Naiad" "LTR/Gypsy" [7] "DNA/TcMar-Tc1" "LINE/CR1" "LTR/Pao" [10] "DNA/hAT" "Satellite" "DNA/PIF-Harbinger" [13] "DNA/TcMar-Tc4" "rRNA" "LTR/Ngaro" [16] "DNA/TcMar-Pogo" "DNA/TcMar-Mariner" "DNA/MULE-MuDR" [19] "snRNA" "SINE/tRNA-RTE" "DNA/hAT-Ac" [22] "LINE/RTE-BovB" "ARTEFACT" "SINE/tRNA-Deu-L2" [25] "DNA/TcMar-ISRm11"